Ananda Homes Salesforce Org — Technical Review

Repo: `anandahomesfullsbox`, branch `Praveen` @ `0d9034a` (2026-10-10) **Default org:** `anandasbox` (drive-innovation-6570--anandasbox) **Reviewer stance:** blunt, line-cited. Everything below was read from source or queried from the sandbox; items I could not execute are marked *(not executed)*.

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0. Executive verdict

Overall grade: D+. There is real engineering talent in this codebase (the newer post-sales services and the config-driven approval/document/report engines are above average for a Salesforce services shop). It is wrapped in a process that cancels most of it out.

The five things a team leader needs to know, in order:

1. **The git repo is not the org, and the org is not the repo.** 182 of 300 Apex classes in git (the entire post-sales rebuild, dynamic approvals, dynamic forms, document engine, reports, dashboards) do not exist in `anandasbox`, and 31 custom objects they depend on (`Demands__c`, `NOC__c`, `Approval_Configuration__c`, …) are missing there too. Conversely, 138 classes in the org (mobile APIs, CP portal, `BulkRaiseDemandController`, `ReceiptController`, …) are not in git. None of the four other connected orgs holds the post-sales code either. All 37 flows in git are absent from the sandbox. **Nobody can currently say which org this repository describes.**
2. **Git HEAD does not compile.** `AgentDetailsAPI.cls:47` has an extra `}` introduced by today's merge commit `47ef1cb` ("Merge branch 'Praveen' into Tejshwini"). The org copy is fine. The next full deploy from this branch fails.
3. **The sandbox is in poor health.** Org-wide Apex coverage is **21%** (production needs 75%). A full `RunLocalTests` run I executed today: 1,055 methods, **154 failed, 3 classes fail to compile** ("Field is not writeable: Payment_schedule__c.Name"), 20 of 132 classes never completed. The last comparable run before mine was June 2026 (37 failures). Nobody is running the suite.
4. **The money path, as written in git, has several defects that would mis-state customer balances**: receipt allocation evaluates a Boolean that is never set (null → exception) *(not executed)*; the open-demand status filter uses two values that do not exist in the picklist; refund ceilings count pending and rejected receipts as cash; cloned bookings on unit swap get their payment schedules twice. Three BRD gates the code claims to enforce (receipt locking, one-pending-receipt, milestone sequencing) are commented out.
5. **Security is not enforced in code anywhere.** Across 137 production classes: 0 `WITH SECURITY_ENFORCED`, 0 `WITH USER_MODE`, 0 `stripInaccessible`. 29 classes are `without sharing`, 29 more have no sharing keyword at all. A live telephony API key and auth key are committed in `click2callController.cls:4-5`. At least 13 `Database.query` sites are injectable with client-supplied input. One VF controller mints public, password-less file links for any file Id it finds in rendered HTML.

Salesforce coding standards: **partially followed in the newest ~40 classes, not followed in the rest.** The project has a PMD ruleset that catches every category of violation present here. It has never been run.

1. Inventory

Area	Count	Notes
Apex classes	300 (137 prod, 163 test)	30.6k prod LOC, 27.0k test LOC
Apex triggers	21	one per object in git; sandbox has 3 on standard `Lead`
Flows	37	31 Active, 3 Draft, 3 Obsolete; 15 on `Booking__c`
LWC	37	29.5k lines js+html; no Jest, no ESLint, no package.json
Aura	19	9.9k lines; still actively authored (Click2Call added Sep 2026)
Visualforce	47 pages	22 hard-coded per-customer/per-bank PDF templates, ~20 Sites boilerplate
Custom objects	69 `__c`, 4 `__mdt`, 1 `__e`	2,285 fields; `Booking__c` alone has 388
Custom metadata records	244	227 country codes
Permission sets	1 real	profiles not in source; access model unreproducible
API versions	classes 56–68, triggers 61–65	`sfdx-project.json` says 62; governs nothing
Contributors	2 humans + bot	Tejaswini 28 commits (two identities), Praveen 6, bot 16

Business domain. Real-estate CRM: pre-sales (Lead__c, Related_Source__c, Site_Visit__c, Follow_up__c, round-robin, channel partners, telephony), sales (Quote__c, Cost Sheet, Plot__c unit blocking), post-sales (Booking__c → Payment_Schedule__c → Demands__c → Receipt__c → Receipt_Line_Item__c, credit/debit notes, refunds, interest, NOC, swaps, cancellations), plus a set of metadata-driven engines (approvals, forms, documents, email, reports, dashboards, field mapping).

2. Source of truth: repo vs org (CRITICAL, process)

Evidence gathered from `anandasbox` today:

Check	Result
Classes in git but not in org	**182** (all post-sales services, all Dynamic* / Approval* / Report* / Dashboard* / Email* / Document* engines, all their tests, `AgentDetailsAPI`, `HMACSignatureService`, …)
Classes in org but not in git	**138** (`All*API`, `CP*`, `*MobileAPI`, `BulkRaiseDemandController`, `ReceiptController`, `GenerateSaleDeed`, `ManualSharingClass`, `PaymentScheduleBatch`, `LeadTriggerHandlerStandard`, …)

Custom objects in git but not in org	**31** incl. `Demands__c`, `Advance_Payment__c`, `Payment_Schedule_Revision__c`, `NOC__c`, `First_Applicant_Change__c`, all 7 config objects
Flows in git present in org	**0 of 37**
Triggers	git has `ConstructionMilestoneTrigger`, `DemandTrigger`, `FirstApplicantChangeTrigger`, `NOCTrigger`, `PaymentScheduleRevisionTrigger`, `ReceiptLineItemTrigger` (not in org); org has `AdditionalChargeReceiptTrigger`, `AdditionalChargesTrigger`, `DemandRaisedTrigger`, `ChannelPartnerTrigger`, `LeadTriggerStandardLead`, `LeadTriggerHandlerStandard` (not in git)
`LeadTrigger` body	org (modified 2026-10-09) has an extra profile in the admin list and the Pre-Sales record-type assignment **commented out**; git has neither change. Deploying git would revert an org hotfix.
Other connected orgs (`orchid`, `orchidprod`, `mantri`, `jainshouse`)	none contain the post-sales classes

What this means: the post-sales/config-framework code (roughly 60% of the repo by volume) was developed in some org that is not authenticated on this machine, or was authored locally and never deployed. The "chore: retrieve from DEFAULT" bot commits (16 of 50) overwrite git from an org whose identity is a GitHub secret. Commit `de54671` ("restore rebalanced ProrataInvariantServiceTest reverted by sandbox retrieve") and the deletion of `EmailSenderReceiptOwnerTest.cls` in retrieve `6f2da61` prove the bot has already destroyed hand-written code at least twice.

The org is the master; git is a lossy backup of an unknown org. Until this is fixed, every other finding in this document is a finding about a file, not about a system.

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3. Triggers

21 triggers in git. One trigger per object is respected in git. The sandbox violates it with **three triggers on standard Lead** (`SLeadTrigger`, `LeadTriggerStandardLead`, `LeadTriggerHandlerStandard` — the last is a trigger *named like a handler class* with an empty body).

Trigger	Pattern	Verdict
`LeadTrigger` (443 lines)	Logic inline, all 7 contexts, user SOQL at L10, 9 hard-coded profile names L19-21, ~200 lines commented out, swallowed DML L329/347, recursion guard `Utility.runLeadTrigger` checked at L4 but **never set false anywhere**	**F**
`RelatedSourceTrigger` (140 lines)	Logic inline; `@future` callout per insert; inserts	D

	`LeadShare` on standard Lead	
`MasterPaymentScheduleTrigger`	SOQL inside `for (Trigger.new)` L17-21; disabled by label `Enable_Master_Payment_Schedule_Trigger = false`	D (and off)
`SiteVisitTrigger`, `BlockTrigger`, `UnitBlockHistoryTrigger`	Inline but bulk-safe	C
`ProjectTrigger`	100% commented body	dead
`ReceiptTrigger`	Delegates; **before-insert/update gate calls commented out** L12-15	C (BRD 5.12-5.15 off)
`ReceiptLineItemTrigger`	**Body fully commented** L8	dead (BRD pointer 4 off)
`BookingTrigger`, `QuoteTrigger`, `UserTrigger`, `FollowUpTrigger`, `DemandTrigger`, `NOCTrigger`, `ConstructionMilestoneTrigger`, `PaymentSchedule*`, `FirstApplicantChangeTrigger`, `AdditionalChargeTrigger`	Thin trigger → service class, bulk-safe	B

Feature flags are Custom Labels compared as strings with three different casings (`'TRUE'`, `'True'`, `'true'`). `Enable_Quote_Trigger` has zero references. The standard is a hierarchy Custom Setting or `__mdt` with a checkbox.

Recursion control: eight new services expose a `public static Boolean bypass` and callers save/restore it — correct. The legacy lead domain has none; re-entry paths exist via `RelatedSourceHandler:369`, `FollowUpHandler:81/105`, `SiteVisitTrigger:48`, `QuoteApprovalTriggerHandler:102`.

Flow/trigger overlap: `Booking__c` has a trigger plus 15 flows (11 active, 5 after-update with undefined order, 2 that update `$Record` in after-save and re-fire the stack). `Receipt__c`, `Demands__c`, `Lead__c`, `Follow_up__c` each have both a trigger and record-triggered flows with no documented ownership. **Zero fault connectors** across all 37 flows.

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4. Apex classes — critical findings

Consolidated from four domain reviews; each item was read at the cited line. "Would fail" items marked *(not executed)* were traced, not run.

4.1 Compile / deploy blockers

- **`AgentDetailsAPI.cls:47`** — extra `}` from today's merge. Git HEAD will not deploy.

- Sandbox: `SaleAgreementControllerTest`, `RaiseDemandControllerTest`, `SwappingUnitControllerTest` fail to compile ("Field is not writeable: Payment_schedule__c.Name" — someone changed Name to auto-number without fixing tests).

4.2 Money integrity (post-sales, as written in git)

- **`FinancialTransactionController.cls:477-478, 543, 863, 883`** — `wrapper.useAdvances/useCredits` assignments are commented out; the fields are `null` Booleans evaluated in `if (wrapper.useAdvances && …)`. Apex throws `NullPointerException` on a null Boolean condition, `allocateReceipt` has no try/catch, so **every receipt approval would throw** *(not executed)*. The suite hides it: `FinancialTransactionControllerTest.cls:1486-1575` wraps every allocate call in `catch (Exception e) { /* ok */ }`.
- **`PostSalesConstants.cls:26-28`** — `DEMAND_OPEN_STATUSES = {'Open','Partial'}`; the `Demands__c.Status__c` picklist is `Raised | Emailed | Reminder Sent | Partially Paid | Paid | Overdue | Cancelled | On Hold | Termination Sent | Unit Unblocked`. `getOpenDemands` (L127) and `getOpenDemandsForAllocation` (L1111) can never return rows.
- **`FinancialTransactionController.cls:45-48, 1071-1088`** — refund ceiling derived from `Total_Receipt_Amount__c`, a roll-up with no status filter (counts Pending and Rejected receipts); refund inserted with `Status__c='Approved'` hard-coded.
- **`BookingSwapHandler.cls:181`** — `insert newBooking` fires `Booking_After_Create_Handler` (generates schedules + advance receipt), then the handler clones the old schedules/receipts on top. Duplicate schedules, phantom receipt.
- **`Credit_Note_Financial_Recalc.flow`** L43-94 — resets `Available_Amount__c = Credit_Amount__c` on every save of an Approved credit note; partially-applied credits re-inflate.
- **Two cancellation statuses.** Flow writes `'Cancelled'`; Apex everywhere uses `'Cancellation'` (`PostSalesConstants:41`, `BookingStageValidationService:66/75/81`, `InterestCalculationBatch:47`, `BulkDemandController:173…`). Flow-cancelled bookings keep accruing interest and reminders.
- **`DemandReminderService.cls:503-527`** — auto unit-unblock writes `Status__c='Cancellation'`, which `BookingStageValidationService` rejects; demand flags already committed with no savepoint → half-committed state, unit never released.
- `ReceiptAllocationService.cls:34-35` and `BookingFinancialRecalcService.cls:30` compare to literal `'Approved'`/`'Rejected'` while the picklist also contains `'Approved by Finance Team'` / `'Rejected by Finance Team'`.
- `BookingController.cls:364` — approval process name contains a literal **TAB**; `:408` submits to the wrong process (copy-paste).
- `BulkDemandController.cls:698-700` — trusts client-supplied `demandAmount`.
- `CreditNoteApplicationService` never reduces `Demands__c.Pending_Amount__c`; credit notes have no collection effect.
- Four different definitions of `Demands__c.Pending_Amount__c` across creators (`BookingFinancialRecalcService:217`, `PaymentScheduleViewController:378`, `BulkDemandController:690`, `ProrataDemandService:135` leaves it null).
- `NumberTOWordConvertion.cls:14-15` — `setScale(0)` rounds HALF_EVEN before splitting paise; amounts ≥ .50 print wrong words on customer documents *(not executed)*.

4.3 Security

- **`click2callController.cls:4-8`** — Knowlarity API key, auth key, DIDs committed. Rotate and move to Named Credential.
- **`CustomLookupGlobalController.cls:93-148`** — `@AuraEnabled`, `without sharing`, concatenates `query`, `objectName`, `filterField`, `searchString` raw. Any authenticated user can read any object.
- **`DynamicDocumentPDFController.cls:300-453`** — inner `without sharing` class inserts `ContentDistribution` (no password, no expiry, download allowed) for every 068/069 Id found in template JSON **or in rendered record HTML**. Anyone who can put a shepherd URL in a rich-text field gets a public link to any file in the org.
- **`DynamicFormExecutionController.cls:238-306`** — `saveFormWithChildren` inserts any object/fields the browser names; no `isCreateable`. Same shape at `DynamicApprovalController.cls:1009-1047` (update any field on any record).
- **`EmailSenderController.cls:253-330, 598, 623, 637, 691`** — client JSON drives WHERE clause, operator, ORDER BY direction, VF page to render, and recipients. Render-any-page-and-email-it primitive.
- **`ReportDataService.cls:6`** / **`DashboardDataService.cls:24,126`** — full config JSON accepted from client; visibility rules only applied in the list UI. `dateRange`/`number` filter values concatenated raw (`ReportDataService:960-970`).
- **`DynamicDocumentRenderer.cls:288, 436-439, 538, 890-896`** — record values into HTML unescaped, rendered with `escape="false"` → stored XSS on the VF domain.
- `LeadMergeController.cls:67-157` — merge + hard delete of any `Lead__c` by Id with no access check, 5 DML no savepoint. `LightningFileUploadHandler.cls:17-19` deletes any ContentDocument by Id in system mode.
- `GRE_Lead_Controller.cls:302`, `UserAvailabilityController.cls:25-43`, `UserStatusController.cls:13` — `update` on client-supplied sObjects in system mode (mass assignment).
- `SendLeadDetailsToPulseTeleSystem.cls:27` — hard-coded AWS endpoint, no auth, callout per record, all errors swallowed.
- `HMACSignatureService` and `PayloadMaskingService` are well-written and **unused**; `LeadPushAPI`/`LogPushAPI` rely on OAuth only.
- Repo-wide: **0** `WITH SECURITY_ENFORCED`, **0** `USER_MODE`, **0** `stripInaccessible`; 40 describe checks total, none on the arbitrary-object endpoints above. All 7 config objects are OWD Public Read/Write.

4.4 Correctness (pre-sales)

- **`RelatedSourceHandler.cls:152-169`** — duplicate check keys on `PhoneProject__c` / `Secondary_Phone_Project__c`, both **formula fields** (`IF(NOT(ISBLANK(Phone__c)),Phone__c,'')`). Formula fields are null on `Trigger.new` in before-insert, so `keySet` is empty and the method returns at L166. **Insert-time dedupe is a no-op** *(not executed; 2-minute anonymous Apex confirms)*. The re-open/re-engage flow depends on it.
- `GRE_Lead_Controller.cls:170-177` — writes a `Lead__c` Id into `SLead__c` (lookup to standard `Lead`) → DML type error.
- `pushToSalesController.cls:100,106` — filters `SLead__c` with a `Lead__c` Id → always empty → guards never fire.
- `RelatedSourceHandler.cls:3-6, 362-376` — class-static accumulators survive a failed DML; stale `deleteList` can delete unrelated leads on the next insert in the same transaction.

- `UnitBlockExpiryScheduler.cls:20-21` — only revokes blocks expiring *today*; one missed run leaves units Blocked forever.
- `HomeNotificationController.cls:26,34` — filters on `'Reschedule'`, `'Missed'`; picklist has neither. `LeadTrigger:150` branch on `'New'` is dead (not a picklist value). `'New sales enquiry'` vs `'New Sales Enquiry'` both written.
- `BulkReassignmenController` (674 lines, misspelled) — dead, targets standard Lead, undefined bind `:valuedate` L209, double `WHERE`.

4.5 Governor limits

- `InterestCalculationBatch` batch size 50 × 2–3 SOQL/booking (`:63-75`) → >100 SOQL per execute once ~34 bookings in a chunk have overdue schedules.
- `DemandReminderService.cls:576→779` — SOQL per e-mail inside 200-row loops, ×6 processors; PDF render per demand.
- `ReceiptAllocationService.cls:47` — `allocateReceipt` per receipt (~10 SOQL, ~6 DML each); `Receipt_Financial_Recalc` flow fires on every save with no change filter → ~24 SOQL per single approval.
- `ApprovalActionHandler` invocable: SOQL/DML/`sendEmail` per record per action.
- `DocumentViewerController.cls:135-140, 367, 470` — queries `Markup` of up to 1,000 ApexPages into heap; template JSON loaded twice per template.
- `DynamicApprovalService.hasField` (`:359-365`) → `Schema.getGlobalDescribe()` ~15× per submission.
- `SendLeadDetailsToPulseTeleSystem:30-62` — callout per record in a `@future`.

4.6 Standards scorecard

Standard	Status	Evidence
Thin trigger / handler pattern	**Mixed**	13 of 21 delegate; `LeadTrigger` 443 inline lines
One trigger per object	Git yes, **org no**	3 triggers on `Lead` in sandbox
Bulkification	**Mixed**	New services good; 12 SOQL/DML-in-loop sites remain
Recursion control	**Partial**	8 services with `bypass`; lead domain none; `runLeadTrigger` never set
Sharing declared	**No**	29 `without`, 29 none, 0 `inherited`
CRUD/FLS enforced	**No**	0 uses of any platform mechanism
SOQL injection safe	**No**	13 injectable sites, 12 config-trust sites
No hard-coded IDs	**Mostly**	1 record type Id in `Record_Type_Appartment` flow; profile/RT names by label everywhere
No secrets in code	**No**	`click2callController`
Exception handling	**Poor**	186 `catch { /* ok */ }` in tests; dozens of `System.debug` catches in prod
No `System.debug` in prod	**No**	278 in production classes
No `Test.isRunningTest()`	**No**	8 classes

in prod		
Naming conventions	**Inconsistent**	lowercase classes (`followUpController`, `click2callController`…), PascalCase methods, misspellings (`BulkReassignmen`, `SimplyfyFilesCntrl`, `Intrest`, `Appartment`)
Dead/commented code	**Heavy**	~600 lines commented in lead domain alone; 5 dead classes; 15 Sites `_Old` duplicates; `HelloWorld` scaffold
ApexDoc / authorship	**Boilerplate**	40× `@author System`, 6× `@author Your Name`, "added by claude" ×6
Decimal for money	**Yes**	no `Double` found
Custom labels for UI strings	**No**	0 `@salesforce/label` imports in LWC

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5. Tests (grade D+)

Git suite. 163 test classes; **28 have zero asserts**; `System.assert(true)` ×19 in `DynamicApprovalControllerTest`, ×17 in `BookingControllerTest` (its only asserts). The flagship `FinancialTransactionControllerTest` is 1,712 lines, 76 methods, **6 asserts, 97 swallowing catches**. `FinancialTransactionControllerTest2` has 0 asserts and `return`s early when setup fails. `Test`/`Test2`/`Test3` split is explicitly for line-coverage chasing (headers say so). 15 classes use `SeeAllData=true` (all Sites boilerplate). Bulk (200-record) tests exist in 4 classes; `LeadTriggerTest` has none. 35 tests query `Profile WHERE Name = '…'` and 7 silently `return` if the profile is missing. `postSalesTestDataFactory` is used by ~17 classes; everywhere else a 40-line `Plot__c` block is copy-pasted. `click2callController` (does a callout) has no test.

Honest estimate: ~35-40% of test classes would catch a regression; the rest is coverage padding.

Sandbox run (today, `RunLocalTests`, job `707HE00000KJOjg`):

Metric	Value
Org-wide coverage	**21%**
Methods enqueued / completed / failed	1,055 / 927 / **154** (+3 compile failures)
Classes enqueued / completed	132 / 112
Top failure causes	`INVALID_OR_NULL_FOR_RESTRICTED_PICKLIST` ×73, `CANNOT_INSERT_UPDATE_ACTIVATE_ENTITY` ×30 (trigger errors inside test inserts), `STRING_TOO_LONG` ×16
Worst classes	`AllSiteVisitAPITest` 28, `AnnouncementImageUploadControllerTest` 19,

	`ProjectDetailAPITest` 16, `BulkReassignmenController_Test` 13, `HomeDashboardAPITest` 12
Lead-domain failures in classes that *are* in git	`SLeadTriggerHandlerTest` 3, `RelatedSourceHandlerTest` 2, `SLeadTriggerTest` 2, `LeadTriggerTest` 1, `followUpControllerTest` 1

The sandbox cannot be promoted to production in its current state.

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6. Flows (grade D)

- 37 flows, **0 fault connectors**, including 19 Apex invocable calls and 7 DML elements.
- `Booking__c`: 15 flows (11 active). Five after-save update flows with undefined order. `Record_Type_Appartment` and `Update_booking_Stage` update `$Record` in after-save → re-fire the whole stack.
- **`Record_Type_Appartment`** sets `RecordTypeId = '0129I000001MJZhQAO'` (hard-coded, org-specific) on every new Booking with no criteria.
- Five flows have **no entry criteria** (`Lead_Assignment`, `Lead_Assignment_to_Sales_User`, `Follow_Ups_Assignment`, `Send_Demand_Email`, `Record_Type_Appartment`).
- `Send_Receipt_Email` fires on `Approval_Status__c = Approved` without `IsChanged` → likely re-sends on every edit.
- Three near-identical names (`Approval_Notification`, `Approval_Notification_Flow`, `Approval_Notifications_Flow`) on three objects; six flows share the label "Recalculate Booking Financials"; two labelled "Copy 1 of Send Notification"; one described as "Test".
- 3 Draft + 3 Obsolete flows committed. `Submit_For_Approval` targets an approval process not in source.

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7. LWC / Aura (grade C)

- **No `package.json`, no ESLint, no Jest, no `__tests__`.** `.forceignore` excludes test/lint files that were never created.
- Monoliths: `documentDesigner.js` 1,567 lines, `formConfiguratorBuilder.js` 1,357, `financialTransactionManager.js` 1,315 (27 Apex imports, 4 transaction types, ~60 tracked fields), `GRE_LeadCreationController.js` 989.
- `@track` on primitives in 32/35 components (CI skeleton comment says it was removed).
- Swallowed errors: `bulkRaiseDemand.js:92,217,220`, `documentDesigner.js:998,1379-1398`, `financialTransactionManager.js:204-268` (8 loaders → empty tables, no toast), `dashboardConfigurator.js:322-326,870`.
- 87 `console.log` in 19 LWC files; `newLeadComponentController.js:96,106` logs the full lead payload (phone, email) to the browser console.
- 469 hard-coded UI strings, **0** `@salesforce/label`.
- `getEmailHistory` → `EmailSenderController.cls:943` sorts a wrapper that does not implement `Comparable`; its test catches the exception and passes.
- Aura is live, not legacy: `Click2Call` added 2026-09-03; 13 quick-action components whose QuickAction metadata is not in source; 6 Aura bundles have zero references.

- `helloWorld` LWC + `HelloWorld` class + test are CI scaffold shipped to a customer org, exposed in App Builder.

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8. Metadata hygiene (grade D+)

- **Profiles, QuickActions, ApprovalProcesses, CustomApplications, CustomNotificationTypes, Named Credentials, Groups, Roles, Sharing Rules are not in source.** The repo cannot rebuild an org.
- Feature flags in Custom Labels (see §3). `Org_Wide_Address` label is a personal Gmail. `intergrationType` typo in an API name.
- Field typos propagated into code: `Intrest_Percentage__c`, `Intrest_Rate__c`, `Actual_Booking_Amout__c`, `Anniverssary_Date1__c`, `Avaliable__c`, `Unit_Type_Intrested__c`. 49 fields end in `1__c` (`Project__c`+`Project1__c`, `Booking_Amount__c`+`Booking_Amount1__c`, three Aadhaar fields).
- **Dual lead model**: standard `Lead` and `Lead__c` coexist; `SLead__c` (→Lead), `CLead__c` (→Lead__c), `New_Lead__c` (→Lead__c) all labelled "Lead" on `Booking__c`, `Follow_up__c`, `Quote__c`, `Site_Visit__c`. Three confirmed bugs above come from confusing them.
- 22 hard-coded Visualforce PDF templates (~8.7k lines), six named after **individual customers** (`ARJUN_SINGH_AGREEMENT_OF_SALE` …), 12 per-bank TPA pages, most with no controller, while a real `DynamicDocument*` engine exists alongside. `DocumentViewerController` glues them together by regex-scanning `ApexPage.Markup` at runtime.
- ~20 Sites/Communities boilerplate pages and 5 `_Old` controller triplets with no Site/Network metadata in source.
- `Drizzle_Logo.png` and `Prosper_Logo.png` are 70 bytes (broken).
- `Project_Name_Mappings__mdt` has zero records but is read by `Utility.getAllProjectNameMappings()`.

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9. CI/CD and process (grade D)

- All three workflows are `workflow_dispatch` only. **Nothing runs on push or PR.**
- `03-deploy-to-salesforce.yml:6`: *"Just deploys. No scan, no validation, no tests."* `--test-level NoTestRun` at L362/378/395, including for the `PRODUCTION` target (Salesforce will reject it).
- `pmd-ruleset.xml` exists, is never executed, and lists exactly the rules this code violates (`AvoidSoqlInLoops`, `ApexCRUDViolation`, `ApexSOQLInjection`, `AvoidDebugStatements`, `ApexUnitTestClassShouldHaveAsserts`, `AvoidHardcodingId`, `EmptyCatchBlock`).
- Delta deploy diffs `HEAD~1..HEAD` (one commit), not since last deploy. Filename/`name:` are swapped between the 02 and 03 workflows. Salesforce CLI unpinned in deploy, pinned in retrieve.
- Retrieve bot defaults to committing straight to the current branch with `[skip ci]`; it has reverted human commits at least twice (see §2).

- `origin/main` has 6 commits, latest subject `"m"`; every working branch is 42-45 commits ahead. One PR ever merged (today). No CODEOWNERS, no PR template, no branch protection evidence.
- `README.md` is the unmodified skeleton ("my-sfdc-project"). `destructiveChanges.xml` is the empty template.
- Human commit messages are good (conventional, specific, reference QA items). That is the one bright spot.

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10. What is done well

Credit where due; this is not a hopeless codebase.

- `BookingFinancialRecalcService`: idempotent full recompute with change detection and bypass. The right pattern, and it is why the triple flow-fire does not double-count.
- `PaymentScheduleRevisionService.stageAndSubmit`: trial `update` inside a savepoint to reuse trigger invariants as the single validator, then rollback. Clever and correct.
- `BookingSwapHandler`: generic clone with old→new Id registry, stale-reference fix-up, bypass restored in `finally`.
- `ProrataInvariantService` / `PaymentScheduleEditorController` share one invariant definition and say so.
- Static `bypass` flags saved and restored by every caller in the new services.
- `ApprovalActionHandler` / `ApprovalGenericStatusHandler`: per-action try/catch isolation, Chatter diagnostics so admins see why an action did not fire; describe-validated field names.
- `ReportDataService` / `DashboardDataService`: regex-whitelisted identifiers, whitelisted aggregations and operators (only the date/number branches slipped).
- `EmailSenderInvocable`: builds all messages, one `Messaging.sendEmail`.
- `LogPushAPI`: idempotent on `call_uuid`, consistent error helper, body truncated before logging. Best class in the lead domain.
- `BulkLeadReassignmentController` + `TempAssignmentExpiryBatch`: bulkified, audited, share removal scoped to the exact pair.
- `UnitBlockingController`, `UserAvailabilityHistoryHandler`, `NOCApprovalSyncService`, `BookingStageRemarkService`, `MilestoneStatusRemarkService`: clean, small, bulk-safe.
- `HMACSignatureService` (constant-time compare) and `PayloadMaskingService`: production quality, waiting to be wired in.
- Test classes `FinancialTransactionControllerTest3` (45 asserts), `DemandReminderServiceTest` (59), `ApprovalActionHandlerTest` (41), `BulkLeadReassignmentController_Test` (37), `LeadTriggerTest` (22) are real tests.
- Comments in the new code explain *why* (BRD sections, BUG ids, platform quirks). Most findings in this report were found because of them.

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11. Grades

Area	Grade

Source control / source of truth	**F**
CI/CD	D
Triggers	D
Pre-sales Apex (legacy core)	D
Post-sales Apex (new services)	C+ (B for design, D for the money defects)
Config-framework Apex	C (A for ambition, D for security)
Security (CRUD/FLS/injection/secrets)	**F**
Tests	D+
Flows	D
LWC / Aura	C
Metadata hygiene	D+
Overall	**D+**

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12. Recommended order of work

This week (stop the bleeding)

1. Fix `AgentDetailsAPI.cls:47`. Add a validate-only (`--dry-run`) deploy on every PR so this cannot recur.
2. Rotate the Knowlarity keys; move them to a Named Credential; purge from git history.
3. Decide the source of truth. Identify the "DEFAULT" org, retrieve *everything* from it into a fresh branch, diff against `anandasbox`, and write down which org owns which module. Disable the retrieve bot's commit-to-current-branch default.
4. Run anonymous Apex to confirm the `useAdvances` null and the formula-field dedupe; fix both; un-swallow the catches in `FinancialTransactionControllerTest`.
5. Fix `DEMAND_OPEN_STATUSES`, the `Cancelled`/`Cancellation` fork, and the refund roll-up filter.

This month

1. Enable PMD in CI with the existing ruleset; fail the build on `ApexSOQLInjection`, `ApexCRUDViolation`, `AvoidSoqlInLoops`, `ApexUnitTestClassShouldHaveAsserts`. Expect hundreds of hits; triage by severity.
2. Add `WITH USER_MODE` (or `stripInaccessible`) to every `@AuraEnabled` reader on the generic engines; add `isCreateable/isUpdateable` to the generic writers; fix the 13 injectable query sites.
3. Remove the public `ContentDistribution` minting; use `ContentVersion.VersionData` base64 only.
4. Refactor `LeadTrigger` into a handler; replace `Utility.runLeadTrigger` with a real guard; replace label flags with a `Trigger_Setting__mdt`.
5. Add fault paths to all 19 Apex-calling flows; delete the 6 Draft/Obsolete flows; consolidate the five after-update Booking flows and set `triggerOrder`.
6. Delete dead code: `BulkReassignmenController`, `Cost_SheetController`, `leadToSFController`, all `_Old` Sites classes/pages/tests, `HelloWorld*`, `ProjectTrigger`, `FormulaBuilderController`.

This quarter

1. Get the sandbox to green: fix the 3 compile failures, the 73 restricted-picklist failures, then raise coverage toward 75% with asserting tests. Add 200-record bulk tests for every trigger.
2. Bring profiles (or convert to permission sets), quick actions, approval processes and apps into source.
3. Add ESLint + Jest to LWC; remove `console.log`; move UI strings to labels.
4. Retire the 22 hard-coded VF PDF templates into `Document_Template__c`.
5. Reconcile the dual lead model (`SLead__c`/`CLead__c`/`New_Lead__c`) or at minimum document it and add a validation rule that one is populated.

Method: 300 classes, 21 triggers, 37 flows, 37 LWC, 19 Aura, 47 VF pages and 74 object folders were read; four parallel domain reviews plus repo-wide static scans; live queries against `anandasbox` (class/trigger/object/flow inventory, `LeadTrigger` body diff, org-wide coverage, a full `RunLocalTests` run). No metadata was deployed and no org data was changed.